

ABHISHEK SINGH

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with 1+ year of hands-on experience designing and building production-grade LLM-powered systems and Generative AI applications. Specialized in Retrieval-Augmented Generation (RAG) pipeline architecture, large language model integration, prompt engineering, and NLP — with practical experience in model fine-tuning using PyTorch and HuggingFace Trainer API. Proficient in the complete RAG stack including text chunking strategies, embedding models, FAISS vector similarity search, context window management, and LangChain memory chains. Passionate about building intelligent AI systems that solve real business problems.

TECHNICAL SKILLS

AI & LLM: Generative AI · LLM Integration · LangChain · RAG Pipeline Design · FAISS Vector Store · OpenAI API · Prompt Engineering · Semantic Search · HuggingFace · NLP · Text Chunking · Embedding Models · Vector Similarity Search · Context Window Management · Memory Chains

Model Training: PyTorch · HuggingFace Trainer API · Model Fine-Tuning · GPT-2 · Text Classification · Named Entity Recognition (NER) · TensorFlow

Programming & Data: Python · FastAPI · Flask · RESTful APIs · Pydantic · Pandas · NumPy

Databases: MongoDB · SQL · Vector Databases · FAISS

DevOps & Cloud: AWS (EC2, S3) · Docker · Nginx · Gunicorn · Linux/Ubuntu · CI/CD Pipelines · Cloudflare R2

Tools: Git · GitHub · Postman · VS Code · Jupyter Notebook

PROFESSIONAL EXPERIENCE

AI Engineer | Flamingo Infinite Technologies Pvt. Ltd., Andheri, Mumbai

Jun 2025 – Present

- Architected and built an end-to-end RAG pipeline using LangChain, FAISS vector store, and OpenAI API — implementing advanced text chunking strategies, embedding models, and vector similarity search to convert unstructured business documents into accurate, queryable knowledge bases.
- Engineered LangChain memory chains and context window management logic to maintain multi-turn conversation history, enabling the AI system to deliver contextually accurate responses across complex, extended user interactions.
- Developed LLM-powered analytics modules using OpenAI API and LangChain that automatically interpret uploaded datasets and generate AI-driven strategic insights, enabling non-technical business users to extract value from complex data conversationally.
- Fine-tuned GPT-2 using HuggingFace Trainer API and PyTorch for text classification and named entity recognition (NER) tasks on domain-specific enterprise data, improving model accuracy for business-specific language patterns.
- Designed and optimised structured prompt templates and response formatting logic using advanced prompt engineering techniques to improve AI output consistency and reduce hallucinations across all enterprise use cases.

Technologies: Python · OpenAI API · LangChain · RAG · FAISS · NLP · Prompt Engineering · HuggingFace · PyTorch · GPT-2 · Flask · AWS

AI Engineer (Trainee) | GM Infotech (Acquired by Flamingo Infinite Technologies Pvt. Ltd.), Andheri, Mumbai

Jan 2025 – Jun 2025

- Developed AI-driven automation workflows integrating OpenAI API and LangChain into backend services, enabling intelligent and context-aware application behaviour across development and production environments.
- Built NLP-powered microservices using Python and FastAPI for text classification, sentiment analysis, and named entity extraction — with full dependency isolation for consistent and reproducible inference results.
- Developed an LLM-powered Q&A module leveraging LangChain memory chains, FAISS vector similarity search, and embedding models to enable semantic search over HR documents, reducing manual query resolution effort for the HR team.
- Implemented text chunking strategies and context window management techniques to optimize RAG pipeline performance, ensuring accurate retrieval and minimal context loss across large document collections.

Technologies: Python · FastAPI · OpenAI API · LangChain · FAISS · HuggingFace · NLP · MongoDB · Prompt Engineering · Pandas · NumPy

PROJECTS

AI-Powered Document Intelligence & Data Insights Platform | [GitHub](#)

- Architected a full RAG pipeline using LangChain and FAISS with custom text chunking strategies, embedding models, and vector similarity search — enabling semantic Q&A and document summarization over mixed document types including PDF, DOCX, and TXT.
- Implemented LangChain memory chains and context window management to support multi-turn conversations, allowing users to ask follow-up questions and receive contextually accurate AI-generated responses across extended sessions.
- Built a dataset insights engine that automatically detects document domain, applies NLP techniques for entity extraction, and presents AI-interpreted findings through interactive Plotly.js visualizations alongside natural language explanations.
- Designed a modular backend architecture using FastAPI with separate service layers for document ingestion, vector indexing, LLM orchestration, and response formatting — ensuring clean separation of concerns and easy extensibility.

Technologies: Python · LangChain · FAISS · OpenAI API · Prompt Engineering · NLP · FastAPI · Flask · Socket.IO · React.js · Plotly.js · MongoDB · Pandas · NumPy

EDUCATION

Master of Computer Applications (MCA)

Aditya Institute of Management Studies & Research (Mumbai University), Borivali, Mumbai | 2023 – 2025

Bachelor of Computer Applications (BCA)

Global Education Trust, Tilak Maharashtra Vidyapeeth University, Bandar, Mumbai | 2020 – 2023

CERTIFICATIONS

- AWS Fundamentals – Coursera
- Database Management System – NPTEL
- Software Development Life Cycle (SDLC) – NPTEL